

Summary - Heterogeneous Multi Agent

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1. Overall Summary

The paper addresses the challenge of dynamically optimizing cache storage and content fetching in cognitive radio networks (CRNs) due to unpredictable content popularity and primary user band occupation. Existing deep reinforcement learning (DRL) approaches often treat the system as a single agent or homogeneous agents, failing to account for the distinct objectives of users (minimum delivery delay) and cache servers (maximum revenue/reputation). To overcome this, the authors propose a novel heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework. This approach models users and Secondary Base Stations (SBSs) as two different types of agents, each with its own action and observation space, learning to cooperate and compete for mutual benefits. The MADDPG allows SBSs to optimize the cached fraction of content in a continuous action space, offering more flexibility than discrete actions. Utilizing centralized training for critics and decentralized execution for actors, the system reduces communication overhead. Numerical simulations demonstrate that the proposed heterogeneous MADDPG significantly outperforms other single-agent or homogeneous DRL methods, including DDPG, PPO, and TD3, by achieving a lower average delivery delay for users and higher cache efficiency for SBSs, leading to an overall system reward improvement of nearly 40%.

2. Main Idea

The central concept is to propose a heterogeneous multi-agent deep deterministic policy gradient (MADDPG) approach for distributed caching in cognitive radio networks (CRNs). This approach models both users and secondary base stations (SBSs) as distinct types of agents that learn to cooperate and compete, with the goal of simultaneously reducing user content delivery delay and enhancing SBS cache efficiency.

3. Problem Being Solved

The paper addresses several limitations and research gaps in previous systems for distributed caching in Cognitive Radio Networks (CRNs):

- 1) Unpredictable Content Popularity: The dynamic and unpredictable nature of content popularity makes it difficult to optimize cache storage and content fetching strategies.
- 2) Inefficient Centralized Learning: Previous centralized learning approaches (e.g., LSTM-based DQN) incur high communication overhead.
- 3) Homogeneous Agent Modeling: Many DRL approaches model the entire system as a single global agent or treat all cache nodes as homogeneous agents. This overlooks the distinct and often conflicting objectives of different entities: users aim for minimum delivery delay, while cache nodes (SBSs) aim to maximize reputation or revenue by storing popular content.
- 4) Discrete Action Space Limitations: Some methods (like DQN) operate only in discrete action spaces, meaning SBSs can only evict or retain whole content, which limits flexible cache usage.
- 5) Lack of Cooperation/Competition Handling: Previous systems often do not effectively model or optimize for the complex mixed cooperative-competitive environment where both homogeneous and heterogeneous agents interact.

4. Assumptions

- The overall contents consist of F files, and the file size s_f is independent identically distributed (i.i.d) of Pareto distribution with an average value of 5.
- The contents are modeled to be static, but the number of requests $q_{k,f}$ is time variant with respect to users.
- Each file can be divided into fragments by using a rateless code like Raptor codes.
- Users can only retrieve one fragment of a file from an SBS and the rest from the PBS, rather than employing multiple SBSs to cooperatively relay fragments.
- SBSs are geographically closer to users than PBS, resulting in a better transmission rate and less delivery delay.
- The slot duration T_0 is long enough that all content requests of users can be served within a time slot.
- The transmission rate of SBSs is faster than that of PBS ($r_{n,k} > r_{0,k}$).

5. Limitations

- The simulation environment is relatively small ($N=5$ SBSs, $K=3$ users, $F=50$ files), and scalability to much larger networks might introduce new challenges.
- The specific parameters used (e.g., incentive price, replacement coefficient) are fixed in the simulation, and their optimal values or sensitivity to real-world variations are not fully explored.
- The assumption that users can only retrieve one fragment from an SBS and the rest from PBS simplifies the problem but might not reflect all real-world scenarios of cooperative caching.

6. Results and Conclusion

The numerical simulations demonstrate the effectiveness and superiority of the proposed heterogeneous MADDPG approach: 1) Convergence: The rewards for both users and SBSs converge along with training episodes, proving the stability and effectiveness of the MADDPG algorithm. 2) Impact of Location and Popularity: SBSs farther from users show worse rewards and adapt their caching strategy to prioritize "hottest" contents, sacrificing less popular ones, which is a rational behavior to attract more users. More popular content is consistently stored by a larger fraction. 3) Impact of Replacement Cost: Increasing the replacement cost (β) shifts the caching strategy towards prioritizing more popular contents, especially when cache space is limited. 4) Superiority over Homogeneous MADRL: The heterogeneous MADDPG significantly outperforms homogeneous MADDPG approaches (where only SBSs or only users learn) in terms of overall system reward, achieving nearly a 40% improvement. This is attributed to its ability to stimulate users to fetch from SBSs (lower delay) and prompt SBSs to optimize storage (better revenue) simultaneously. 5) Outperformance against Other DRL Algorithms: The proposed MADDPG also achieves higher global rewards compared to other actor-critic based off-policy DRL algorithms like DDPG, Proximal Policy Optimization (PPO), and Twin Delayed DDPG (TD3). In conclusion, the heterogeneous MADDPG effectively balances the mixed cooperative-competitive relationships between users and SBSs, leading to simultaneous optimization of content storage and delivery delay, thereby enhancing the overall system reward and efficiency in CRNs.

7. Real World Impact

1) Enhanced Wireless Network Performance: By reducing content delivery delay and improving cache efficiency, the research can lead to faster and more reliable content access for users in CRNs. 2) Efficient Spectrum Utilization: Optimizing caching strategies in CRNs helps free up spectrum resources for other users, addressing spectrum scarcity. 3) Edge AI and Distributed Systems: The multi-agent learning approach can be applied to other distributed systems where autonomous agents with different objectives need to cooperate and compete, such as edge computing, IoT networks, and smart city infrastructure. 4) Dynamic Resource Management: The ability to dynamically adapt caching strategies based on content popularity and user demand has implications for dynamic resource allocation in various communication networks. 5) Improved User Experience: Lower delivery delays directly translate to a better quality of experience for end-users consuming content.

8. Graphs and Figures Analysis

Figure 1: System model

This figure illustrates the architecture of the Cognitive Radio Network (CRN) system. It shows a Primary Base Station (PBS) at the center, surrounded by multiple Secondary Base Stations (SBSs), and various Users. The diagram indicates two types of access: primary access from PBS (dashed line) and secondary access from SBSs (solid line). It also depicts the multi-agent learning framework, showing two types of agents: K Agents of Users and N Agents of SBSs. Each agent type has an Actor network (π) and a Critic network (Q), with their respective states (S), actions (A), and rewards (R). The "Training" phase involves a global state (S_{All}) and global action (A_{All}) for the Critic, while the "Execution" phase shows decentralized actions.

Table I: Neural Network Configuration of MADDPG

This table details the architecture of the neural networks used for the Actor and Critic components of the MADDPG algorithm for both User and SBS agents. For the Actor, it specifies an input layer of size F (for requested times or content popularity) with ReLU activation, two hidden layers with 300 neurons each and ReLU activation, and an output layer of size F (for fetching vector or cached fraction) with Sigmoid activation. For the Critic, it specifies an input layer of size $2(N+K)F$ (for global state-action pair) with ReLU activation, two hidden layers with 1024 neurons each and ReLU activation, and a scalar output (1) for the policy gradient value.

Figure 2: Locations and rewards of heterogeneous agents

This figure presents three sub-plots related to agent locations and their rewards during training. (a) Position Distribution: A scatter plot showing the spatial distribution of the PBS, five SBSs (labeled SBS1 to SBS5), and three Users (labeled User1 to User3) in a 2D coordinate system (x, y in meters). It visually indicates that SBSs 2 and 4 are located farther from the cluster of users compared to other SBSs. (b) Users' reward: A line graph showing the reward convergence for each of the three users (User 1, User 2, User 3) over 300,000 training episodes. All user rewards converge to negative values, with User 2 consistently having the lowest (most negative) reward, indicating higher delivery delay for User 2. (c) SBSs' reward: A line graph showing the reward convergence for each of the five SBSs (SBS 1 to SBS 5) over 300,000 training episodes. The rewards for all SBSs converge to positive values. SBSs 2 and 4 show lower (but still positive) rewards compared to SBSs 1, 3, and 5, which aligns with their farther geographic location from users.

Figure 3: Contents popularity vs. cached fraction

This figure consists of two sub-plots illustrating how content popularity and replacement cost affect caching strategy. (a) Impact by location of SBS: A line graph showing the average cached fraction (λ) as a function of normalized content popularity for each of the five SBSs. The curves generally show that more popular content has a higher cached fraction. SBSs 2 and 4 exhibit slightly different curves, indicating they cache a larger fraction of less popular content compared to other SBSs, which is a strategy to compensate for their less favorable geographic location. (b) Impact by factor β : A line graph showing the average cached fraction (λ) as a function of normalized content popularity for the 1st SBS under different replacement cost factors ($\beta = 0.5, 0.8, 1.0, 1.2$). As β increases (higher replacement cost), the curves shift from being more "up-concave" to "down-convex," demonstrating that SBSs tend to cache a larger fraction of more popular content and a smaller fraction of less popular content, prioritizing hot content when replacement costs are high or cache space is rare.

Figure 4: Comparison with homogeneous MADDPG

This figure presents two sub-plots comparing the overall rewards of the proposed heterogeneous MADDPG with homogeneous MADDPG variants. (a) Users' overall reward: A line graph showing the sum of users' rewards over training episodes for three scenarios: Heter-MADDPG (users and SBSs), Homo-MADDPG (SBSs Only), and Homo-MADDPG (users Only). The heterogeneous MADDPG consistently achieves a higher (less negative) overall reward for users, indicating better performance in terms of delivery delay. (b) SBSs' overall reward: A line graph showing the sum of SBSs' rewards over training episodes for the same three scenarios. The heterogeneous MADDPG consistently achieves a higher overall reward for SBSs, indicating better cache efficiency and revenue.

Figure 5: Performance comparison with other DRL

This line graph compares the global reward (sum of all users' and SBSs' rewards) of the proposed heterogeneous MADDPG with other state-of-the-art DRL algorithms: DDPG, PPO, and TD3, over 300,000 training episodes. The MADDPG curve shows a significantly higher and more stable global reward compared to all other DRL algorithms, demonstrating its superior performance in optimizing the system's overall objectives (delivery delay and cache efficiency).